

Лабораторная работа № 4

Бондарь Т. В.

Российский университет дружбы народов, Москва, Россия

Информация

- Бондарь Татьяна Владимировна
- НКАбд-01-24, студ. билет №1132246711
- Российский университет дружбы народов
- https://github.com/tvbondar/study_2024-2025_os-intro

Получение навыков правильной работы с репозиториями git.

1. Выполнить работу для тестового репозитория.
2. Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

Gitflow Workflow опубликована и популяризована Винсентом Дриссенем. Gitflow Workflow предполагает выстраивание строгой модели ветвления с учётом выпуска проекта. Данная модель отлично подходит для организации рабочего процесса на основе релизов. Работа по модели Gitflow включает создание отдельной ветки для исправлений ошибок в рабочей среде.

Семантическое версионирование описывается в манифесте семантического версионирования. Кратко его можно описать следующим образом: Версия задаётся в виде кортежа МАЖОРНАЯ_ВЕРСИЯ.МИНОРНАЯ_ВЕРСИЯ.ПАТЧ. Номер версии следует увеличивать: МАЖОРНУЮ версию, когда сделаны обратно несовместимые изменения API. МИНОРНУЮ версию, когда вы добавляете новую функциональность, не нарушая обратной совместимости. ПАТЧ-версию, когда вы делаете обратно совместимые исправления. Дополнительные обозначения для пререлизных и билд-метаданных возможны как дополнения к МАЖОРНАЯ.МИНОРНАЯ.ПАТЧ формату.

1. Устанавливаем git-flow

```
chris@chrisfedor:~$ cd /tmp
chris@chrisfedor:~/tmp$ wget --no-check-certificate -q https://raw.githubusercontent.com/petervanderdoes/gitflow/develop/contrib/gitflow-installer.sh
chris@chrisfedor:~/tmp$ chmod +x gitflow-installer.sh
chris@chrisfedor:~/tmp$ sudo ./gitflow-installer.sh install stable
## git-flow do make installer ##
Installing git-flow to /usr/local/bin
Cloning repo from github to gitflow
Cloning into 'gitflow'...
remote: Enumerating objects: 4278, done.
remote: Total 4278 (delta 0), reused 0 (delta 0), pack-reused 4278 (from 1)
Receiving objects: 100% (4278/4278), 1.74 MiB | 1.41 MiB/s, done.
Resolving deltas: 100% (4053/4053), done.
```

Рис. 1: Установка git-flow

2. Устанавливаем Node.js

```
root@kali:~/# sudo apt install nodejs
Last metadata expiration check: 8:04:19 ago on Mon 24 Feb 2025 08:23:23 AM EST.
Dependencies resolved.
=====
Package                               Architecture          Version               Repository            Size
-----
Installing:
nodejs                                x86_64                1:20.18.2-2.fc40     updates               52 k
Installing dependencies:
nodejs-devel                          x86_64                1:20.18.2-2.fc40     updates               16 M
Installing weak dependencies:
nodejs-devel-docs                     x86_64                1:20.18.2-2.fc40     updates               6.8 M
=====
```

Рис. 2: Установка Node.js


```

dnf install -y vim
Last metadata expiration check: 8:06:07 ago on Mon 24 Feb 2025 08:23:23 AM EST.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
-----
Installing:
vim                    noarch            9.0.8-1.fc40     updates           2.0 M
Transaction Summary
-----
Install 1 Package

```

3. Для работы с Node.js добавим каталог с исполняемыми файлами, устанавливаемыми yarn, в переменную PATH.

```
tybondary@fedora:~$ npm setup
Appended new lines to /home/tybondary/.bashrc

Next configuration changes were made:
export NPM_HOME="/home/tybondary/.local/share/npm"
case "$PATH" in
  *"$NPM_HOME") ;;
  *) export PATH="$NPM_HOME:$PATH" ;;
esac

To start using npm, run:
source /home/tybondary/.bashrc
tybondary@fedora:~$ source ~/.bashrc
bash: /home/tybondary/.bashrc: No such file or directory
tybondary@fedora:~$ source ~/.bashrc
tybondary@fedora:~$ npm add -g commitizen

Update available: 0.0.0 > 10.4.1
Change log: https://github.com/npm/npm/releases/tag/v10.4.1
Run "npm add -g npm" to update.
Follow @npmjs for updates: https://twitter.com/npmjs

npm WARN deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +151
Progress: resolved 151, reused 0, downloaded 151, added 151, done

/home/tybondary/.local/share/npm/global/3:
  commitizen 1.1.1

Done in 9.3s
```

Рис. 3: Настройка Node.js

4. Настраиваем commitizen, standard-changelog.

```
tsdonan@fedora:~$ npm add -g commitizen

  Update available! 0.0.0 -> 10.0.1.
  Changelog: https://github.com/commitizen/commitizen/releases/tag/v10.0.1
  Run "npm add -g commitizen" to update.
  Follow @commitizen for updates: https://twitter.com/commitizen

npm WARN deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +151
Progress: resolved 151, reused 0, downloaded 151, added 151, done
/home/tsdonan/.local/share/npm/global/
commitizen 10.0.1
Done in 5.3s
```

Рис. 4: commitizen

```
root@kali:~# npm add -g standard-changelog
npm WARN deprecated @babel/plugin-proposal-private-property-in-object@7.21.0: This proposal has been merged into the ECMAScript standard and is now shipped in Node.js 18.0.0
Packages: +39
Progress: resolved 189, reused 133, downloaded 39, added 39, done
/home/rykondar/.local/share/npm/global/5:
standard-changelog 0.0.0
Done: 1h 3.7s
root@kali:~#
```

Рис. 5: standard-changelog

5. Создаем репозиторий на GitHub. Для примера назовём его git-extended. Делаем первый коммит и выкладываем на github:

```
tbondar@fedora:~/git-extended$ git add .
tbondar@fedora:~/git-extended$ git commit -m "first commit"
[main 66b2881] first commit
1 file changed, 1 deletion(-)
tbondar@fedora:~/git-extended$ git remote add origin git@github.com:tbondar/git-extended.git
error: remote origin already exists.
tbondar@fedora:~/git-extended$ git remote -v
origin  git@github.com:tbondar/git-extended.git (fetch)
origin  git@github.com:tbondar/git-extended.git (push)
tbondar@fedora:~/git-extended$ git remote add git@github.com:tbondar/git-extended.git
usage: git remote add [-options] <name> <url>

    -f, --[no-]fetch          fetch the remote branches
    --[no-]tags              report all tags and associated objects when fetching
                           or do not fetch any tag at all (--no-tags)
    -t, --[no-]track <branch>      branch(es) to track
    -m, --[no-]master <branch>      master branch
    --[no-]mirror[=(<push|fetch>)] set up remote as a mirror to push to or fetch from

tbondar@fedora:~/git-extended$ git push -u origin master
error: src refspec master does not match any
error: failed to push some refs to 'git@github.com:tbondar/git-extended.git'
tbondar@fedora:~/git-extended$ git push -u master
fatal: 'master' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
tbondar@fedora:~/git-extended$ git push -u
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3) | 248 bytes | 248.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:tbondar/git-extended.git
  630096a..16b2881  main -> main
branch 'main' set up to track 'origin/main'.
tbondar@fedora:~/git-extended$
```

Рис. 6: git-extended

6. Конфигурация общепринятых коммитов. Для этого добавим в файл package.json команду для формирования коммитов:

```
/home/tvbondar/git-extended/package.json
{
  "name": "git-extended",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:tvbondar/git-extended.git",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "Tatiana Bondar sinopahfox@yandex.ru",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рис. 7: Конфигурация коммитов

7. Добавляем файлы, выполняем коммит, отправляем на Github.

```
tvbondar@fedora:~/git-extended$ git add .
tvbondar@fedora:~/git-extended$ git cz
Parsing JSON at /home/tvbondar/git-extended/package.json for commitizen config failed:

tvbondar@fedora:~/git-extended$ mc

tvbondar@fedora:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing:
> revert: Reverts a previous commit
  feat:   A new feature
  fix:    A bug fix
  docs:   Documentation only changes
  style:  Changes that do not affect the meaning of the code (white-space, formatting, missing semi-colons, etc)
  refactor: A code change that neither fixes a bug nor adds a feature
  perf:   A code change that improves performance
(Move up and down to reveal more choices)
tvbondar@fedora:~/git-extended$ git add .
tvbondar@fedora:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: feat:   A new feature
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 94 chars):
  (19) change_package.json
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[main c7366dd] feat: change_package.json
1 file changed, 18 insertions(+)
 create mode 100644 package.json
tvbondar@fedora:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 573 bytes | 573.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:tvbondar/git-extended.git
 563b201..c7366dd main -> main
tvbondar@fedora:~/git-extended$
```

8. Инициализируем git-flow. Префикс для ярлыков установим в v. Проверьте, что Вы на ветке develop: Загрузите весь репозиторий в хранилище: Установите внешнюю ветку как вышестоящую для этой ветки: Создадим релиз с версией 1.0.0

```
tvbondar@fedora:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- main
Branch name for production releases: [main]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/tvbondar/git-extended/.git/hooks]
tvbondar@fedora:~/git-extended$
```

Рис. 9: Часть 1


```
tvbondar@fedora:~/git-extended$ git branch
* develop
  main
tvbondar@fedora:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/tvbondar/git-extended/pull/new/develop
remote:
To github.com:tvbondar/git-extended.git
 * [new branch]      develop -> develop
tvbondar@fedora:~/git-extended$
```

Рис. 10: Часть 2

```
tvbondar@fedora:~/git-extended$ git branch --set-upstream-to=origin/develop develop
branch 'develop' set up to track 'origin/develop'.
tvbondar@fedora:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'
```

Рис. 11: Часть 3

9. Создадим журнал изменений Добавим журнал изменений в индекс Зальём релизную ветку в основную ветку Отправим данные на github Создадим релиз на github. Для этого будем использовать утилиты работы с github:

```
tvbondar@fedora:~/git-extended$ yarn global add standard-changelog
yarn global v1.22.22
[1/4] Resolving packages...
[2/4] Fetching packages...
[3/4] Linking dependencies...
[4/4] Building fresh packages...
success Installed "standard-changelog@6.0.0" with binaries:
  - standard-changelog
Done in 8.44s.
tvbondar@fedora:~/git-extended$ npx standard-changelog
Need to install the following packages:
standard-changelog@6.0.0
Ok to proceed? (y) y

✓ created CHANGELOG.md
✓ output changes to CHANGELOG.md
tvbondar@fedora:~/git-extended$
```

Рис. 12: Часть 1

```

tybondar@fedora: ~/git-extended$ npx standard-changelog
Need to install the following packages:
standard-changelog@6.0.0
Ok to proceed? (y) y

✓ created CHANGELOG.md
✓ output changes to CHANGELOG.md
tybondar@fedora: ~/git-extended$ ls
CHANGELOG.md  package.json  README.md
tybondar@fedora: ~/git-extended$ standart-changelog --first-release
bash: standart-changelog: command not found...
tybondar@fedora: ~/git-extended$ git add CHANGELOG.md
tybondar@fedora: ~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 121e622] chore(site): add changelog
1 file changed, 10 insertions(+)
create mode 100644 CHANGELOG.md
tybondar@fedora: ~/git-extended$ git flow release finish 1.0.0
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 10 ++++++++
1 file changed, 10 insertions(+)
create mode 100644 CHANGELOG.md
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)
fatal: no tag message?
Fatal: Tagging failed. Please run finish again to retry.
tybondar@fedora: ~/git-extended$ git flow release finish 1.0.0
Branches 'main' and 'origin/main' have diverged.
And local branch 'main' is ahead of 'origin/main'.
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 10 ++++++++
1 file changed, 10 insertions(+)
create mode 100644 CHANGELOG.md
Deleted branch release/1.0.0 (was 121e622).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'main'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

tybondar@fedora: ~/git-extended$

```

Рис. 13: Часть 2

```
tvbondar@fedora:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 691 bytes | 691.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:tvbondar/git-extended.git
   c7366dd..159ff27  develop -> develop
   c7366dd..f33b639  main -> main
tvbondar@fedora:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 169 bytes | 169.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:tvbondar/git-extended.git
 * [new tag]         v1.0.0 -> v1.0.0
tvbondar@fedora:~/git-extended$
```

Рис. 14: Часть 3

```
tvbondar@fedora:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md  
https://github.com/tvbondar/git-extended/releases/tag/v1.0.0  
tvbondar@fedora:~/git-extended$
```

Рис. 15: Часть 4

10. Создадим ветку для новой функциональности. Следующим шагом следует объединить ветку `feature_branch` с `develop`.

```
tvbondar@fedora:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

tvbondar@fedora:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was 159ff27).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

tvbondar@fedora:~/git-extended$
```

Рис. 16: Создание и слияние веток

11. Создадим релиз с версией 1.2.3. Обновим номер версии в файле package.json. Установим её в 1.2.3.

```
tvbondar@fedora:~/git-extended$ git flow release start 1.2.3
Switched to a new branch 'release/1.2.3'

Summary of actions:
- A new branch 'release/1.2.3' was created, based on 'develop'
- You are now on branch 'release/1.2.3'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.2.3'

tvbondar@fedora:~/git-extended$
```

Рис. 17: Релиз v1.2.3.


```
package.json  [-M--] 19 L:[ 1+ 2 3/ 19] *(47 / 441b) 0034 0x022
{
  "name": "git-extended",
  "version": "1.2.3",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:tvbondar/git-extended.git",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "Tatiana Bondar sinopahfox@yandex.ru",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
<----->"path": "cz-conventional-changelog"
<----->}
    }
  }
}
```

Рис. 18: Изменение package.json

12. Создадим журнал изменений Добавим журнал изменений в индекс Зальём релизную ветку в основную ветку Отправим данные на github Создадим релиз на github с комментарием из журнала изменений:

```
tvbondar@fedora:~/git-extended$ npx standard-changelog  
✓ output changes to CHANGELOG.md  
tvbondar@fedora:~/git-extended$
```

Рис. 19: Часть 1

```

tvbondar@fedora:~/git-extended$ git add CHANGELOG.md
tvbondar@fedora:~/git-extended$ git commit -am 'chore(site): update changelog'
[release/1.2.3 5f6603d] chore(site): update changelog
 2 files changed, 5 insertions(+), 1 deletion(-)
tvbondar@fedora:~/git-extended$ git flow release finish 1.2.3
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Merge made by the 'ort' strategy.
  CHANGELOG.md | 4 ++++
  package.json | 2 +-
  2 files changed, 5 insertions(+), 1 deletion(-)
Already on 'main'
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
  CHANGELOG.md | 4 ++++
  package.json | 2 +-
  2 files changed, 5 insertions(+), 1 deletion(-)
Deleted branch release/1.2.3 (was 5f6603d).

Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'main'
- The release was tagged 'v1.2.3'
- Release tag 'v1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

tvbondar@fedora:~/git-extended$

```

```
tvbondar@fedora:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 759 bytes | 759.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To github.com:tvbondar/git-extended.git
    159ff27..f882a67  develop -> develop
    f33b639..e980d10  main -> main
tvbondar@fedora:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 172 bytes | 172.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:tvbondar/git-extended.git
 * [new tag]          v1.2.3 -> v1.2.3
tvbondar@fedora:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/tvbondar/git-extended/releases/tag/v1.2.3
tvbondar@fedora:~/git-extended$
```

Рис. 21: Часть 3

Мы получили навыки правильной работы с репозиториями git.

Список литературы

Рабочий процесс с Gitflow(электронный ресурс) URL:

<https://yamadharma.github.io/ru/post/2021/04/18/gitflow-workflow/>